

Literature Review — Cross-Dataset Generalization of ML-Based Intrusion Detection

Project: Evaluating and Improving Cross-Dataset Generalization of ML-Based Intrusion Detection Systems **Document:** 02 — Literature Review (focused set of **15 core papers**) **Citation style:** IEEE (numbered)

1. How this review is organised

This review covers **15 core papers** divided into five themes. Each paper is summarised using the same structure — **problem, method, datasets, key results, limitation** — followed by “*what it means for our project*” so the team can answer questions on any single paper.

Theme A — Cross-dataset generalization (the core problem): [1]–[4] **Theme B — Benchmark datasets:** [5]–[7], [14] **Theme C — Metaheuristic / bio-inspired feature selection:** [8]–[12] **Theme D — Domain adaptation & transfer learning:** [13] **Theme E — Evaluation and dataset-quality critique:** [15]

2. Theme A — Cross-Dataset Generalization

[1] Cantone, Marrocco & Bria (2024) — *On the Cross-Dataset Generalization of ML for NIDS*, IEEE Access

- **Problem:** Do ML-based NIDS still work when trained on one network and tested on a different one?
- **Method:** Four classifiers (Decision Tree, Random Forest, KNN, MLP) are trained and tested across **four** datasets (CIC-IDS-2017, CSE-CIC-IDS2018, LycoS-IDS2017, LycoS-Unicas-IDS2018) using a **restricted common feature representation**; the authors also introduce a corrected version of CSE-CIC-IDS2018 and use data visualisation to diagnose failures.
- **Datasets:** CIC-IDS-2017, CSE-CIC-IDS2018, LycoS-IDS2017, LycoS-Unicas-IDS2018.
- **Key results:** (i) near-perfect accuracy when training and testing on the same dataset (~0.99); (ii) when tested cross-dataset, accuracy falls to roughly **random chance** for most source→target pairs; (iii) failures are caused by **dataset-specific artefacts** — identity fields (IP addresses, ports) and collection methodology — not by weak models.
- **Limitation:** It only *diagnoses* the problem. It does **not** propose or optimise a fix — it uses a fixed common feature set and default hyper-parameters.
- **What it means for us:** This is our closest prior work and our baseline. Our project *extends* it by adding a **metaheuristic that selects features for transfer** instead of using a fixed subset, and by using a third structurally different dataset (TON_IoT).

[2] Hossain et al. (2026) — *Assessing Generalisation Capability of ML Models for Intrusion Detection*

- **Problem:** Does high accuracy within one dataset imply reliability on an unseen dataset?

- **Method:** Random Forest, Logistic Regression, and Naïve Bayes evaluated under both same-dataset and cross-dataset settings on **UNSW-NB15 ↔ TON_IoT**.
- **Datasets:** UNSW-NB15 and TON_IoT (two of our three datasets).
- **Key results:** Random Forest reaches **95.08%** accuracy on UNSW-NB15 and **99.79%** on TON_IoT in-dataset, but cross-dataset accuracy drops **below 40%** in both directions — a very large generalisation gap.
- **Limitation:** Purely observational. No feature selection, no optimisation, no mitigation, and no CIC-IDS-2017.
- **What it means for us:** This is almost a subset of our project. It proves the gap exists on UNSW/TON; we add CIC-IDS-2017 and, crucially, attempt to *close* the gap with transfer- oriented optimisation. We must cite and differentiate from it.

[3] Hakim, Uddin & Anis (2026) — *Cross-Domain Generalization Failure in Lightweight IDS for IIoT*

- **Problem:** Do lightweight IDS models transfer to structurally different IIoT networks?
- **Method:** Four lightweight models are trained on one IIoT source and tested (no retraining) on two targets, using a feature representation restricted to attributes common to all three; SHAP-style explainability, plus adversarial-robustness and limited-adaptation probes.
- **Key results:** (i) the best models rely overwhelmingly on **coarse port-category features**, which appear **96–435× more often** in the source attack traffic than in the targets — a **shortcut**, not a genuine attack signature; (ii) the evaluation protocol (how class imbalance is handled) can **reverse** which target looks harder; (iii) adversarial robustness is **unrelated** to cross-network generalisation.
- **Limitation:** Again diagnostic; it does not select features for transfer.
- **What it means for us:** Directly shapes our design — we must **remove port/identity features** and evaluate under **realistic class distributions** to avoid the shortcut trap.

[4] Butt, Hotho & Schlör (2026) — *Evaluating Tabular Representation Learning for NIDS*

- **Problem:** Can learned representations (instead of hand-crafted features) transfer better across networks?
- **Method:** Systematic evaluation of state-of-the-art tabular representation-learning methods (e.g. TabICL), autoencoders, and end-to-end transformers on several NetFlow datasets, with comprehensive hyper-parameter search and cross-dataset transfer experiments.
- **Key results:** Strong **dataset–model dependency** — no method dominates everywhere; cross-dataset transfer can work, but performance varies greatly depending on the source–target pair.
- **Limitation:** Model/representation-centric; no feature selection; no optimisation targeting transfer.
- **What it means for us:** Supports our premise that *what we feed the model* matters as much as the model itself, and warns us to report per-pair results rather than a single average.

3. Theme B — Benchmark Datasets

[5] Sharafaldin, Lashkari & Ghorbani (2018) — CIC-IDS-2017

- **Contribution:** Introduces CIC-IDS-2017 (CICFlowMeter, 78 flow features, 5 days of realistic enterprise traffic, 14 attack types), with a documented testbed and attack schedule.
- **Key facts:** Realistic benign traffic via B-Profile; flows extracted with CICFlowMeter; the most-used NIDS benchmark to date.
- **Known issues (from later work):** identity/port features inflate same-dataset scores; extreme class imbalance; occasional duplicate/NaN/Inf values.
- **What it means for us:** Primary **source** dataset; we use its flow statistics and map them to the shared subspace.

[6] Moustafa & Slay (2015) — UNSW-NB15

- **Contribution:** Introduces UNSW-NB15 (IXIA PerfectStorm generator; Argus + Bro features; 9 attack families + Normal) plus a fixed train/test split.
- **Key facts:** 45-column public split; both binary `label` and 9-class `attack_cat`.
- **What it means for us:** Second dataset; its `ct_*` time-window features are exactly the kind of collection-dependent signal we expect not to transfer.

[7] Moustafa (2021) — TON_IoT

- **Contribution:** Introduces the TON_IoT IoT/IloT datasets collected from a three-layer (edge-fog-cloud) SDN/NFV testbed; the network portion uses raw Zeek (Bro) log fields.
- **Key facts:** 10 attack classes (near-uniformly balanced), heavy categorical/protocol fields.
- **What it means for us:** Third dataset and the most realistic *target* domain (rawest schema); its balanced classes make it ideal for clean macro-metric evaluation.

[14] Ring et al. (2019) — *A Survey of Network-Based Intrusion Detection Data Sets*

- **Contribution:** A comprehensive taxonomy and comparison of NIDS datasets, covering collection environment, feature semantics, labelling method, and reuse.
 - **Key facts:** Shows datasets differ radically in how they were built — which is the root cause of cross-dataset mismatch.
 - **What it means for us:** Justifies why we must build a **common feature subspace** before comparing datasets.
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4. Theme C — Metaheuristic / Bio-Inspired Feature Selection for IDS

[8] Fatima & Ali (2022) — GA wrapper features + ensembles

- **Problem:** Too many features slow detection and hurt accuracy.
- **Method:** Wrapper **Genetic Algorithm (GA)** feature selection combined with stacking and bagging ensembles for **multi-class** detection.
- **Datasets:** UNSW-NB15 and CICDDoS2019.

- **Key results:** GA + stacking improves accuracy, detection rate, and lowers false-alarm rate versus baselines.
- **Limitation:** Fitness is **same-dataset** accuracy — the subset is optimised for the training distribution, so transfer is not guaranteed.
- **What it means for us:** This is the exact pattern we invert: we keep a GA but change its **fitness to cross-dataset performance**.

[9] Suman, Tripathy & Saha (2019) — **NSGA-II unsupervised feature selection**

- **Method:** **NSGA-II** multi-objective optimisation of three information-theoretic measures (mutual information, standard deviation, information gain) for *unsupervised* selection; the Pareto-optimal subsets feed SVM/DT/KNN.
- **Datasets:** KDD-99, NSL-KDD, Kyoto 2006+.
- **Key results:** 99.78% (KDD-99), 99.83% (NSL-KDD 20% test), 99.65% (Kyoto).
- **Limitation:** Legacy datasets; single-dataset evaluation; no transfer objective.
- **What it means for us:** Motivates our optional **multi-objective** variant (cross-dataset F1 vs number of features).

[10] Li (2026) — **MPDGGA**

- **Method:** A multi-population, diversity-guided GA with an information-gain-ratio operator to maintain population diversity when selecting features from high-dimensional traffic.
- **Datasets:** NSL-KDD, UNSW-NB15, and 9 UCI datasets.
- **Key results:** Best accuracy on 10/11 datasets while selecting very few features ($\geq 2.26\%$ of features).
- **Limitation:** Still same-dataset accuracy; no cross-dataset protocol.
- **What it means for us:** Shows modern GA variants can be very compact — useful when our shared subspace is small.

[11] Mojtahedi et al. (2022) — **GA + Whale Optimization hybrid**

- **Method:** Hybrid **Whale Optimization Algorithm (WOA)** + GA feature selection with KNN.
- **Datasets:** KDDCUP1999.
- **Key results:** Better accuracy than prior methods by extracting class-relevant features.
- **Limitation:** Legacy dataset; accuracy-only fitness; single domain.
- **What it means for us:** Demonstrates that **swarm-intelligence hybridisation** (like our planned PSO/WOA) is a legitimate, published design choice.

[12] Yang et al. (2026) — **NSGA-III AutoML IDS**

- **Method:** **NSGA-III** jointly optimises a LightGBM feature-selection threshold and key hyperparameters under three objectives: weighted F1, 99th-percentile inference latency, and model size.
- **Datasets:** CICEVSE2024 and CICIDS2017.
- **Key results:** Competitive F1 with lower latency and smaller models than baselines.
- **Limitation:** Objectives are deployment-efficiency, not generalisation; single-dataset.
- **What it means for us:** Clarifies our distinction — we optimise for **generalisation**, not latency/size — and gives a template for multi-objective optimisation.

5. Theme D — Domain Adaptation / Transfer Learning

[13] Layeghy, Baktashmotlagh & Portmann (2022/23) — DI-NIDS

- **Problem:** Supervised NIDS degrade badly on out-of-distribution networks.
 - **Method:** Adversarial domain adaptation (DANN-style) learns **domain-invariant features** from labelled sources; a One-Class SVM then detects anomalies in the invariant space.
 - **Datasets:** NFv2-CIC-2018 and NFv2-UNSW-NB15.
 - **Key results:** Superior cross-domain performance vs prior domain-adaptation approaches.
 - **Limitation:** Needs aligned (NetFlow v2) feature spaces; unsupervised detection only; requires target-domain data during adaptation.
 - **What it means for us:** Represents the “model-centric transfer” alternative to our “feature-selection transfer” approach; we compare against it conceptually.
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6. Theme E — Evaluation & Dataset-Quality Critique

[15] Vitorino, Silva, Maia & Praça (2024) — Adversarial robustness benchmark / NewCICIDS

- **Method:** Benchmarks decision-tree ensembles (RF, XGB, LGBM, EBM) with constrained adversarial examples on the original **CICIDS2017**, a corrected version (“**NewCICIDS**”), and the more recent HIKARI dataset.
 - **Key results:** The corrected CICIDS2017 yields **better and different** model rankings, showing the original dataset’s flaws materially affect conclusions.
 - **What it means for us:** Justifies (i) cleaning CICIDS2017 (NaN/Inf, duplicate column) and (ii) removing identity/leakage features before any generalisation claim.
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7. Literature Summary Table (for the report/presentation)

#	Title	Author(s)	What it does	Conclusion / relevance
1	On the Cross-Dataset Generalization of ML for NIDS	Cantone et al., 2024	Cross-dataset benchmark over 4 datasets with a common feature set	Accuracy collapses to near chance cross-dataset; diagnosis only
2	Assessing Generalisation Capability of ML Models for ID	Hossain et al., 2026	RF/LR/NB on UNSW-NB15 ↔ TON_IoT	<40% cross-dataset vs 95–99% in-dataset; no mitigation
3	Cross-Domain Generalization Failure in Lightweight IDS	Hakim et al., 2026	Lightweight models across 3 IIoT sets; SHAP	Port-category shortcut; protocol can reverse conclusions
4	Evaluating Tabular Representation Learning for NIDS	Butt et al., 2026	Representation learning + cross-dataset transfer	Transfer depends on source–target pair
5	CIC-IDS-2017	Sharafaldin et al., 2018	Dataset	Primary source dataset
6	UNSW-NB15	Moustafa & Slay, 2015	Dataset	Second dataset
7	TON_IoT network datasets	Moustafa, 2021	Dataset	Third dataset (Zeek features)
8	Metaheuristic classifiers for multiclass ID	Fatima & Ali, 2022	GA feature selection + ensembles	Strong single-dataset results, no transfer
9	Unsupervised FS via NSGA-II	Suman et al., 2019	Multi-objective FS	Pareto subsets; single-dataset
10	MPDGGA	Li, 2026	Multi-population GA FS	Best on 10/11 datasets; single-dataset
11	GA + Whale Optimization FS	Mojtahedi et al., 2022	Hybrid bio-inspired FS	Swarm hybridisation is viable
12	NSGA-III AutoML IDS	Yang et al., 2026	Multi-objective AutoML	Efficiency objectives, not transfer
13	Survey on NIDS Bus Data SBs Benchmark (NewCICIDS)	Imwanchai et al., 2022	Domain-adversarial on serial invariant features	Improves cross-domain, but needs target data
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8. Research Gap (synthesis)

Group	Cross-dataset eval?	Metaheuristic FS?	Optimised for transfer?
Cantone [1], Hossain [2], Hakim [3], Butt [4]	Yes	No	No (diagnosis only)
Metaheuristic FS [8]–[12]	No (single dataset)	Yes	No (same-dataset fitness)
Domain adaptation [13]	Yes	No	No (aligns features; needs target data)
This project	Yes (3 datasets)	Yes (GA, PSO, ...)	Yes (cross-dataset fitness)

Gap: Metaheuristic feature selection for IDS is always optimised for **same-dataset** accuracy, while cross-dataset studies are **diagnostic** or rely on domain adaptation that needs target-domain data. **No published work selects features using a cross-dataset generalisation objective.** Our project fills exactly this gap.

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